

AUDUBON COUNTY AP, IA, USA

WMO: 725498

Lat: 41.700N

Lon: 94.917W

Elev: 1287

StdP: 14.03

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.2	0.0	-13.4	2.8	-3.4	-8.6	3.6	0.8	25.7	25.0	23.3	25.6	3.6	320	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	91.3	76.2	90.0	75.5	87.5	74.4	80.1	88.1	77.9	85.7	76.0	84.1	10.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	150.0	85.6	75.1	138.3	83.3	73.0	128.7	81.7	44.9	87.9	42.3	86.0	40.4	84.0	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	19.5	17.6	-14.8	97.1	5.1	3.4	-18.5	99.5	-21.4	101.5	-24.3	103.4	-28.0	105.8
			-15.3	82.8	4.6	2.0	-18.6	84.3	-21.3	85.4	-23.9	86.6	-27.3	88.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.2	22.1	24.1	37.6	49.4	60.7	70.5	74.1	70.9	64.3	50.9	37.8	25.8
	DBStd	20.77	11.85	12.07	11.99	9.85	8.66	6.40	5.75	5.72	8.43	9.46	10.46	11.28
	HDD50	3390	866	726	416	127	12	0	0	0	4	104	386	749
	HDD65	6619	1331	1146	850	474	190	21	4	14	113	446	816	1214
	CDD50	3084	0	0	32	110	343	616	746	649	434	133	20	1
	CDD65	836	0	0	1	7	55	187	286	198	93	9	0	0
	CDH74	8659	0	0	13	129	597	1916	2987	1892	1002	119	4	0
	CDH80	3034	0	0	1	28	177	673	1169	635	328	23	0	0
	WSAvg	6.9	7.6	7.8	8.4	8.9	7.7	6.4	5.0	4.6	5.5	6.7	7.6	7.3
	PrecAvg	34.4	0.9	1.1	2.2	3.7	5.3	4.8	4.6	4.0	2.8	2.4	1.7	1.0
(15) Precipitation	PrecMax	45.4	1.9	2.5	5.3	7.6	11.9	12.2	10.2	8.9	7.9	7.9	4.3	2.8
	PrecMin	25.7	0.3	0.1	0.1	0.7	1.3	0.8	1.4	1.0	0.7	0.5	0.1	0.0
	PrecStd	6.1	0.4	0.7	1.5	1.9	2.6	2.9	2.4	2.1	1.7	1.7	1.2	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	57.2	75.0	84.5	90.9	93.1	95.5	94.9	91.2	82.4	72.5	57.1
		MCWB	42.9	47.2	59.3	64.5	69.4	73.2	79.6	76.5	73.9	64.4	58.7	50.3
	2%	DB	46.2	50.5	69.6	78.6	84.3	90.3	91.4	90.2	88.0	77.2	64.4	50.3
		MCWB	40.1	44.6	56.9	60.8	69.5	74.0	78.3	75.7	73.5	61.5	55.4	45.0
	5%	DB	41.5	45.3	63.0	72.6	81.1	87.8	90.1	86.3	83.7	72.4	59.3	45.6
		MCWB	36.5	40.2	53.5	57.7	66.0	72.6	77.6	75.7	71.4	60.7	51.7	40.9
	10%	DB	37.1	39.4	56.7	67.9	76.8	83.9	86.4	82.5	80.8	67.8	54.7	41.2
		MCWB	34.1	36.0	49.1	55.4	63.7	70.9	76.1	72.8	68.7	58.3	47.5	37.0
	0.4%	WB	45.1	49.6	63.4	68.0	75.4	80.2	83.6	81.5	78.0	70.2	61.3	53.8
		MCDB	51.4	54.4	70.6	78.3	83.9	88.4	91.3	88.0	87.0	74.9	67.9	56.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.8	44.6	59.0	63.7	71.9	77.1	81.3	78.5	75.5	66.0	57.1	45.9
		MCDB	46.9	50.1	64.6	73.2	81.8	85.5	89.5	86.5	83.7	73.5	63.1	48.6
	5%	WB	37.5	40.0	54.6	60.2	68.5	75.2	78.9	76.6	73.4	62.5	52.7	41.0
		MCDB	41.6	44.9	61.4	69.8	77.5	83.8	86.5	84.3	81.5	69.4	58.6	45.9
	10%	WB	34.4	36.6	49.2	57.0	66.0	73.2	76.8	74.7	70.7	59.0	48.0	37.5
		MCDB	37.6	40.1	57.0	65.9	74.2	81.4	84.5	82.0	77.7	66.2	54.3	41.2
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.2	20.4	23.1	22.0	21.3	21.7	21.8	24.7	22.9	21.2	17.0
		MCDBR	23.4	23.5	29.9	32.1	26.8	24.0	23.1	25.1	27.1	29.5	29.2	23.5
		MCWBR	18.2	17.8	18.6	18.1	14.1	12.2	12.9	14.8	15.2	17.3	19.4	18.5
	5% WB	MCDBR	21.6	22.1	25.4	25.7	22.9	20.4	20.5	21.6	23.4	23.7	26.1	21.2
		MCWBR	17.1	17.2	17.4	16.9	13.9	12.8	13.5	14.6	15.1	16.7	19.0	17.4
(40) Clear-Sky Solar Irradiance	taub		0.284	0.310	0.323	0.378	0.402	0.413	0.429	0.415	0.380	0.334	0.294	0.284
	taud		2.416	2.354	2.401	2.269	2.273	2.296	2.238	2.286	2.340	2.456	2.497	2.437
	Ebn at Noon		277	284	294	282	277	273	268	268	270	272	271	266
	Edh at Noon		25	31	33	41	42	41	43	40	35	28	23	23
(44) All-Sky Solar Radiation	RadAvg		618	936	1205	1521	1807	2004	2035	1750	1451	994	680	515
	RadStd		56	60	113	118	137	172	139	119	100	120	49	46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page